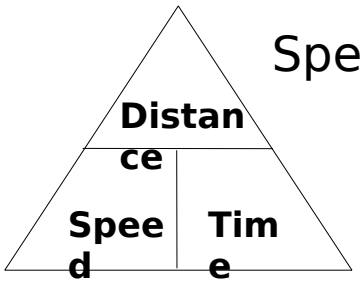
Speed Problems





1. A car travels 400m in 20s. Calculate the average speed of the car in m/s.
2. A cyclist travels 30km in 12 minutes. Calculate the average speed in km/min.
3. A train is travelling at 10m/s. How far does the train travel in 5 seconds?
4. A car is travelling at 15m/s. How long does it take for the car to travel 600m?
5. A bus is travelling at 12m/s. How long does it take for the bus to travel 72km?
6. A trolley travels 20 metres in 1.5 seconds. What is its average speed?
7. A model car travels at 2.5 m/s for 5 seconds. How far does the car travel?
8. Tim drives 400 m to the shop in 20 seconds. Calculate the average speed of the car in km/h. If the speed limit in the area is 70km/h, was Tim speeding?
9. On the weekend, Barry goes for a bike ride. He travels 30km in 20 minutes. Calculate Barry's speed in km/h.
10. Tim's car is travelling at 15m/s. How long does it take him to travel 20 km?
11. Barry walks to Scott's house, a distance of 600m in 5 minutes, but Scott is not home. Barry then walks to John's house, 300m away and it takes him 3 minutes.
    a. What was the total distance Barry walked?

b. How long did Barry's walk take?
c. Calculate Barry's average speed.

12. A plane flies 420km in 1 hour 30 minutes. Find its average speed in km/h.

13. A train travels 510m in a time of 17s. Calculate the average speed of the train in m/s.

14. A boat sails 1.2km due east for a time of 5 minutes. Find the average speed of the boat in m/s.

15. An aeroplane flies with an average speed of 85m/s for 20 s. Calculate the distance the aeroplane flies.

16. A lorry travels along a straight road with a constant speed of 30m/s. After a time of 15 s, how far has the lorry travelled?

17. Australia IV sails in a straight line from one buoy to the next. If the average speed is 4.8km/h and the time taken is 45 minutes, find the distance between the two buoys.

18. Beth rides her horse 3km east and then 4km north. The trip takes her 30 minutes.

a. What is the total distance Beth travelled?
b. What is her average speed in km/h?

19. Amanda drives her car 35km north and then 75km south in a time of 2 hours.

a. What is the total distance she travelled?
b. What is her average speed in km/h?
c. What is her average speed in m/s?

20. A plane flies east at 250km/h for 3 hours and then south at 200km/h for 1 hour.

a. What is the total distance travelled?
b. What is the average speed in km/h?



21. A train travels in a straight line from one station to the next at 40km/h and then returns at 60km/h. If the two stations are 5km apart, how long is the train's journey?

Challenge

Do you remember the fable of the hare and the tortoise? The giant tortoise can reach a speed of about 7.5cm/s. A hare can run as fast as 20m/s. If their race was 1km long, how long was the hare asleep if the race resulted in a tie?